1) Diffie - Hellman Key Exchange Analysis

Given

$P = 23, g = 5$

User A Private key $a = 6$

User B Private key $b = 15$

a) compute Public keys for A and B.

User A public key:

formula : $A = g^a \bmod p$

$A = 5^6 \bmod 23$

$A = 15625 \bmod 23 = 8$

User B Public key

formula $B = g^b \bmod p$

$B = 5^{15} \bmod 23$

$B = 19$

Compute the shared secret key

using userA's private key:

$K = B^a \bmod p$

$K = 19^6 \bmod 23 = 2$

using user B's private key:

$K = A^b \bmod p$

$K = 8^{15} \bmod 23 = 2$

with the attacker.
* the attacker can read or modify messages exchanged between them.

Prevention of MITM Attack

* use digital signatures of authenticate public keys
* use digital certificates issued by a trusted certificate authority (CA).
* use authenticated protocols such as TLS with certificate validation.

Result:-

Diffe-Hellman successfully generates a shared secret. but authentication is necessary to prevent MITM attacks.

Conclusion:-

Diffie-Hellman key exchange helps two users generation a common secret key over an insecure network. In this problem, the shared secret key is 2.. However, a man-in-the-Middle attack can occur. If public keys are not authenticated Digital signatures, digital certificates, and TLS helps prevent

reduces memory usage and communication overhead.

* It can consume less energy in constrained devices
* It Provides strong security with smaller keys than RSA at comparable security levels

→ Power and Memory Consumption

Factor	RSA	ECC
Power consumption	Generally higher	Generally lower
Memory usage	Higher	Lower
Storage	Requires larger keys	Requires smaller keys
Battery Efficiency	Less suitable	More suitable

Result:- ECC is generally preferred for IOT devices because of its smaller keys, lower storage requirements and potential power savings.

Conclusion

ECE is generally more suitable than for resource constrained IoT devices because it uses smaller keys required less Memory, and can reduce Power consumption and communication overhead. Therefore ECE is a preferred choice for IoT Applications requiring efficient and secure communication

(5) ECC vs RSA Comparison

Given

RSA key size = 1024 bits

ECC key size = 160 bits

Devices = IOT devices with limited processing capacity

Computational Cost Comparison

Factor	RSA	ECC
Key Size	1024 bits	160 bits
Computational	Modular exponentiation	Elliptic-curve Point operations
Key generation	can be expensive	Generally efficient
Memory requirement	Higher	Lower
Communication overhead	Higher	Lower

Which is better for IOT devices

ECC is the better choice for the given scenario

Reasons:

* ECC uses smaller keys and requires less storage

Shared secret key = 2

man-in-the middle (MITM) attack

A MITM attacker intercepts and changes the public keys exchanged between user A and user B

User A

Send public key = 8

↓

Attacker (MITM)

Intercepts and replaces public keys

↓

User B

Receives the substitution public key

steps:

* the attacker intercepts A's public key and sends a fake key to B
* the attacker Intercepts B's public key and sends a fake key to A
* A and B unknowingly establish seperate shared secrets